

```
import numpy as np

numbers = np.array([10, 20, 30])

print(numbers)
print(numbers.mean())
```

```
... [10 20 30]
     20.0
```

```
import pandas as pd

data = {
    "Name": ["Arun", "Priya"],
    "Mark": [90, 95]
}

df = pd.DataFrame(data)

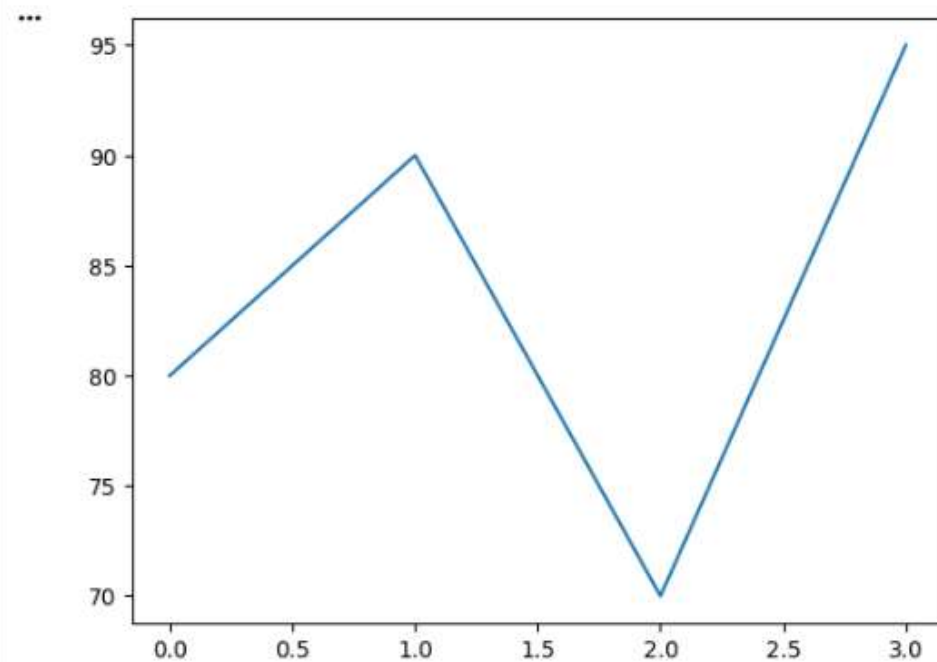
print(df)
```

	Name	Mark
0	Arun	90
1	Priya	95

```
import pandas as pd
data = {
    "Name": ["Madhu", "Arun", "Priya"],
    "Mark": [90, 95, 89]
}
df = pd.DataFrame(data)
print(df)
```

```
...      Name  Mark
0  Madhu    90
1  Arun    95
2  Priya    89
```

```
import matplotlib.pyplot as plt
marks = [80, 90, 70, 95]
plt.plot(marks)
plt.show()
```



```
import numpy as np
marks = np.array([80, 90, 70, 95])
print("Average Mark:", marks.mean())
```

Average Mark: 83.75

```
▶ import numpy as np
numbers = np.array([10, 20, 30])
print(numbers)
```

... [10 20 30]

```
▶ import numpy as np
data = np.array([15, 45, 10, 90, 55])
print("Maximum:", data.max())
print("Minimum:", data.min())
```

... Maximum: 90
Minimum: 10

```
import numpy as np
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print("Addition:", a + b)
```

Addition: [5 7 9]

```
import numpy as np
numbers = np.array([1,2,3,4,5,6])
matrix = numbers.reshape(2,3)
print(matrix)
```

```
[[1 2 3]
 [4 5 6]]
```

```
▶ import pandas as pd
data = {
    "Name": ["Madhushree", "Arun", "Priya", "Kavin"],
    "Mark": [90, 95, 85, 78]
}
df = pd.DataFrame(data)
print(df)
```

```
***
```

	Name	Mark
0	Madhushree	90
1	Arun	95
2	Priya	85
3	Kavin	78

```
▶ import pandas as pd
data = {
    "Name": ["Madhushree", "Arun", "Priya", "Kavin"],
    "Mark": [91, 90, 95, 85]
}
df = pd.DataFrame(data)
print("Highest Mark:")
print(df["Mark"].max())
```

```
*** Highest Mark:
95
```

```
▶ import pandas as pd
data = {
    "Name": ["Madhushree", "Arun", "Priya", "Kavin"],
    "Mark": [96, 90, 95, 85]
}
df = pd.DataFrame(data)
print(df[df["Mark"] > 90])
```

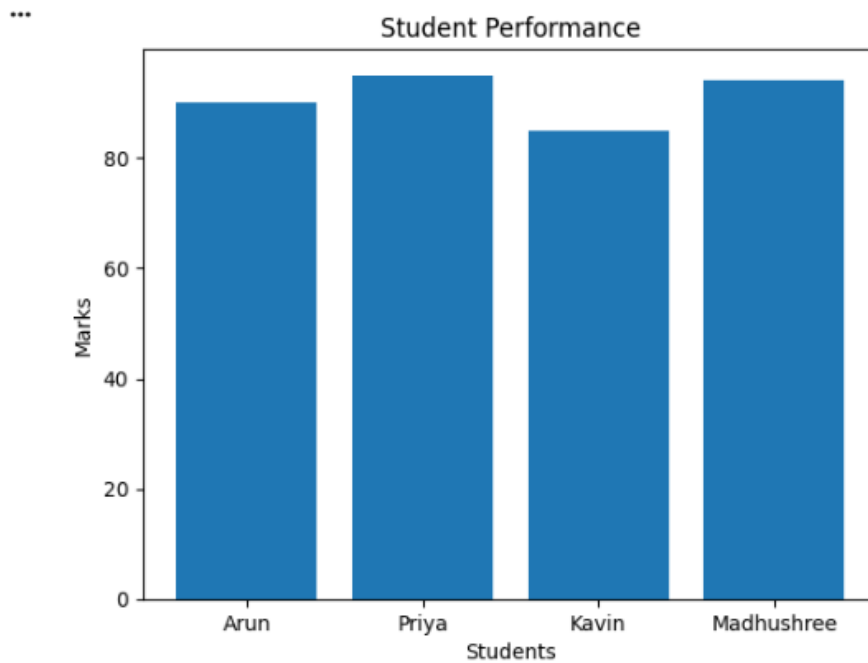
```
***
```

	Name	Mark
0	Madhushree	96
2	Priya	95

```

import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Name": ["Arun", "Priya", "Kavin", "Madhushree"],
    "Mark": [90, 95, 85, 94]
}
df = pd.DataFrame(data)
plt.bar(df["Name"], df["Mark"])
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Performance")
plt.show()

```



```

print("Welcome to the Student Pass Predictor!")

hours_studied = float(input("Enter the number of hours studied: "))
if hours_studied >= 5:
    print("Prediction: PASS! 🎉 Great job studying.")
else:
    print("Prediction: FAIL. 📖 You might want to study a bit more!")

```

```

Welcome to the Student Pass Predictor!
Enter the number of hours studied: 6
Prediction: PASS! 🎉 Great job studying.

```